

GAS DYNAMICS

COMPRESSIBLE FLOW & AEROTHERMODYNAMICS (BGD NOTES)

Isentropic Nozzle Flow, Normal/Oblique Shocks, Fanno/Rayleigh Flow & Propulsion

DOCUMENT SPECIFICATIONS & COURSE INFORMATION

Course Title:	Gas Dynamics & Compressible Flow (BGD Booklet)
Subject Area:	Fluid Mechanics, Thermodynamics & Aerospace Propulsion
Curated For:	Mechanical & Aerospace Engineering Students and Researchers
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SYLLABUS & MODULE OUTLINE // COURSE STRUCTURE

MOD 01	Fundamental Concepts of Compressible Flow Speed of Sound, Mach Number, Stagnation State, Perfect Gas Relations
MOD 02	One-Dimensional Isentropic Flow & Nozzles Area-Velocity Relation, Converging-Diverging (de Laval) Nozzle, Choking
MOD 03	Normal Shock Waves in Supersonic Ducts Rankine-Hugoniot Equations, Total Pressure Losses, Entropy Generation
MOD 04	Oblique Shock Waves & Prandtl-Meyer Expansion Theta-Beta-M Relations, Shock Reflection, Diamond Airfoils in Supersonic Flow
MOD 05	Adiabatic Flow with Friction (Fanno Flow) Fanno Line, Choking due to Wall Friction, Maximum Duct Length
MOD 06	Frictionless Flow with Heat Addition (Rayleigh Flow) Rayleigh Line, Thermal Choking, Maximum Heat Transfer Limits
MOD 07	Aerospace Propulsion & Supersonic Inlets Diffusers, Over/Under Expanded Nozzles, Rocket Exhaust Expansion

CHAPTER 1: FUNDAMENTAL PRINCIPLES & SOUND SPEED

Compressible fluid dynamics governs flows where density variation ($d\rho/\rho$) is substantial, typically occurring when the flow velocity approaches or exceeds the local acoustic velocity ($Mach > 0.3$). Under these regimes, thermodynamics and fluid kinematics become inextricably coupled through state equations.

GOVERNING THERMODYNAMIC RELATIONS FOR COMPRESSIBLE GAS

- **Speed of Sound:** $a = \sqrt{\gamma * R * T}$
- **Mach Number:** $M = V / a$
- **Stagnation T:** $T_0 / T = 1 + [(\gamma - 1) / 2] * M^2$

CHAPTER 2: ONE-DIMENSIONAL ISENTROPIC FLOW & NOZZLES

In isentropic flow, entropy is conserved ($ds = 0$), and total pressure and stagnation temperature remain constant throughout the streamline. The differential area-velocity relationship dictates geometric effects:

$$dA / A = (M^2 - 1) * (dV / V)$$

This fundamental law reveals why supersonic acceleration requires a diverging duct, contrasting directly with intuitive incompressible flow behavior.

KEY ISENTROPIC FLOW RELATIONS (IDEAL GAS, $\gamma = 1.4$)

1. Pressure Ratio: $P_0 / P = [1 + (\gamma - 1)/2 * M^2]^{\gamma / (\gamma - 1)}$
2. Density Ratio: $\rho_0 / \rho = [1 + (\gamma - 1)/2 * M^2]^{1 / (\gamma - 1)}$
3. Critical Pressure: $P^* / P_0 = [2 / (\gamma + 1)]^{\gamma / (\gamma - 1)} = 0.52828$ (for air)
4. Critical Temp: $T^* / T_0 = 2 / (\gamma + 1) = 0.83333$
5. Area-Mach Relation: $A / A^* = (1/M) * [2 / (\gamma + 1)] * (1 + (\gamma - 1)/2 * M^2)^{(\gamma + 1) / (2 * (\gamma - 1))}$
6. Maximum Mass Flux: $\dot{m}_{dot_max} = (P_0 * A^* / \sqrt{T_0}) * \sqrt{\gamma / R} * [2 / (\gamma + 1)]^{(\gamma + 1) / (2 * (\gamma - 1))}$

ENGINEERING HIGHLIGHT: CONVERGING-DIVERGING (DE LAVAL) NOZZLE

- For subsonic entry ($M < 1$), a converging section accelerates the fluid toward the throat.
- Choked flow condition ($M = 1$) is achieved strictly at the minimum cross-sectional area (throat, A^*).
- Further expansion into supersonic velocities ($M > 1$) strictly necessitates a diverging nozzle section.
- If backpressure $P_b >$ design exit pressure P_e , normal shocks or oblique shocks form inside or outside.

CHAPTER 3: NORMAL SHOCK WAVES IN COMPRESSIBLE FLOW

A normal shock wave is an extremely thin discontinuity (order of mean free path $\sim 10^{-5}$ cm) across which flow velocity drops precipitously from supersonic ($M_1 > 1$) to subsonic ($M_2 < 1$), with concomitant jumps in static pressure, temperature, and entropy. Stagnation temperature T_0 is preserved, but total pressure P_0 drops.

RANKINE-HUGONIOT NORMAL SHOCK FORMULATIONS ($\gamma = 1.4$)

1. Downstream Mach: $M_2^2 = [2 + (\gamma - 1)M_1^2] / [2\gamma M_1^2 - (\gamma - 1)]$
2. Static Pressure: $P_2 / P_1 = 1 + [2\gamma / (\gamma + 1)] * (M_1^2 - 1)$
3. Static Temperature: $T_2 / T_1 = [1 + (2\gamma/(\gamma+1))(M_1^2-1)] * [(2 + (\gamma-1)M_1^2)/((\gamma+1)M_1^2)]$
4. Density Jump: $\rho_2/\rho_1 = [(\gamma + 1)M_1^2] / [2 + (\gamma - 1)M_1^2]$
5. Total Pressure Loss: $P_{02} / P_{01} = [(\gamma+1)M_1^2 / (2 + (\gamma-1)M_1^2)]^{(\gamma/(\gamma-1))} * [(\gamma+1)/(\gamma M_1^2)]^{1/(\gamma-1)}$
6. Entropy Generation: $s_2 - s_1 = -R * \ln(P_{02} / P_{01}) > 0$ (Always positive for irreversible shock)

CHAPTER 4: OBLIQUE SHOCKS & PRANDTL-MEYER EXPANSION

When supersonic flow encounters a concave corner or wedge of angle θ , an oblique shock forms at an angle β . The normal component of Mach number governs the shock jump, while the tangential component remains constant across the shock surface.

$$\tan(\theta) = 2 * \cot(\beta) * [(M_1^2 * \sin^2(\beta) - 1) / (M_1^2 * (\gamma + \cos(2\beta)) + 2)]$$

Conversely, supersonic flow expanding around a convex corner generates a Prandtl-Meyer expansion fan, an isentropic expansion where Mach increases and pressure drops smoothly according to the Prandtl-Meyer function $\nu(M)$.

OBLIQUE SHOCK & EXPANSION FAN CRITICAL RULES

- Strong vs. Weak Shock: For a given deflection angle θ , two β angles exist. Weak shock is natural in unconfined flow.
- Shock Detachment: If θ exceeds $\theta_{\max}$ for a given M_1 , the shock detaches and becomes a curved bow shock.
- Prandtl-Meyer Function: $\nu(M) = \sqrt{(\gamma+1)/(\gamma-1)} * \tan^{-1}(\sqrt{(\gamma-1)/(\gamma+1)} * (M^2-1)) - \tan^{-1}(\sqrt{M^2-1})$.
- Expansion Relation: $\nu(M_2) = \nu(M_1) + \theta$ (Flow expands isentropically, total pressure is fully preserved).

CHAPTER 5 & 6: FANNO FLOW (FRICTION) & RAYLEIGH FLOW (HEAT)

FANNO FLOW (Friction, Adiabatic) • Constant Area Duct + Wall Friction • Stagnation Temp $T_0 = \text{Constant}$ • Stagnation Pressure P_0 Decreases • Subsonic flow accelerates toward $M=1$ • Supersonic flow decelerates toward $M=1$ • Friction Choking: $L = L_{\text{max}}$ (Critical length) • $4fL^*/D = (1-M^2)/(\gamma M^2) + ((\gamma+1)/2\gamma) \ln(((\gamma+1)M^2)/2\gamma(1+(1/2)(\gamma-1)M^2))$	RAYLEIGH FLOW (Heat Addition, Frictionless) • Constant Area Duct + Heat Exchange (q) • T_0 Changes: $q = c_p (T_{02} - T_{01})$ • Heating drives Mach toward $M=1$ • Cooling drives Mach away from $M=1$ • Thermal Choking: $q = q_{\text{max}}$ • $T_{02}/T_{0*} = [2(\gamma+1)M^2 (1+(\gamma-1)M^2/2)] / (1+\gamma M^2)^2$ • $P/P^* = (\gamma + 1) / (1 + \gamma M^2)$
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CHAPTER 7: PROPULSION & AEROSPACE NOZZLE REGIMES

In rocket propulsion and supersonic aircraft exhaust nozzles, the relationship between exit pressure (P_e) and ambient atmospheric backpressure (P_b) establishes three distinct operational regimes:

1. Correctly Expanded ($P_e = P_b$): Optimum performance, parallel jet boundary, highest thrust coefficient.
2. Under-Expanded ($P_e > P_b$): Gas continues to expand outside nozzle via expansion fans (high altitude).
3. Over-Expanded ($P_e < P_b$): Ambient pressure pinches the plume, producing oblique shock compression rings and Mach diamonds.

STANDARD GAS DYNAMICS ISENTROPIC & NORMAL SHOCK BENCHMARK TABLE (AIR $\gamma = 1.4$)						
Mach (M_1)	T/T_0	P/P_0	A/A^*	M_2 (Shock)	P_2/P_1	P_{02}/P_{01}
0.50	0.9524	0.8430	1.3398	(Subsonic)	-----	1.00000
1.00 (Sonic)	0.8333	0.5283	1.0000	1.0000	1.000	1.00000
1.50	0.6897	0.2724	1.1762	0.7011	2.458	0.92978
2.00	0.5556	0.1278	1.6875	0.5774	4.500	0.72087
2.50	0.4444	0.0585	2.6367	0.5130	7.125	0.49901
3.00	0.3571	0.0272	4.2346	0.4752	10.333	0.32834